IIT ROORKEE

FREE ONLINE EDUCATION swayam शिक्षित भारत, उन्नत भारत

NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 34: Life Cycle Assessment: Theory & Practice

By-
Prof. Bhola Ram Gurjar
Dept. of Civil Engineering

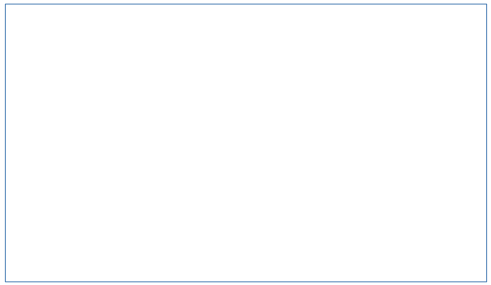


Contents

- ISO (International Organization for Standardization) standards for LCA
  - ISO
  - ISO 14040 series & its description

- Types of LCA as per ISO

- Case study: Environmental benefits of EV vehicles in Milan & Turin (Italy):
  - Study area
  - Goal of study
  - Data
  - Results
  - Conclusion



ISO (International Organization for Standardization)

- ISO is an independent, non-governmental international organization with a membership of 165 national standards bodies
- Develops voluntary, consensus-based, market relevant International Standards that support innovation and provide solutions to global challenges
- In 1947, ISO officially came into existence

ISO 14040 Series

- The ISO 14040 series standards consist Life Cycle Assessment procedure.
- It addresses quantitative assessment methods for the assessment of the environmental aspects of a product or service in its entire life cycle stages.
- ISO 14040 is an overarching standard encompassing all four phases of LCA.
- There are various standards supplementing ISO 14040.

ISO 14040 series standards: Description 1/2

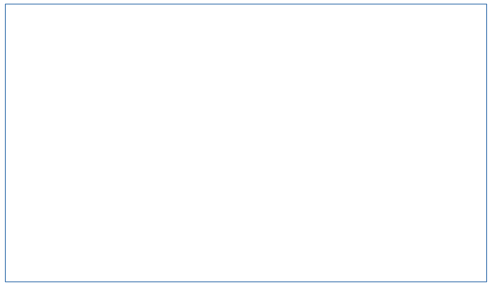
ISO standard	Description
• 14040	General principles, framework and requirements for the LCA of products and services
• 14041	Guidance on determining the goal and scope of a LCA study and for conducting a life cycle inventory
• 14042	Guidance on conducting the LCIA phase of LCA
• 14043	Guidance on the interpretation of results from a LCA study

ISO 14040 series standards: Description 2/2

ISO standard	Description
• 14047	Provides illustrative examples on how to carry out LCIA
• 14048	Information on formatting of data to support LCA
• 14049	Examples that illustrate how to apply the guidance in ISO 14041

Types of LCA

- There are three different types of LCA
    - Conceptual LCA
    - Simplified LCA
    - Detailed LCA
- Used in different ways and each have its strengths and weaknesses
- Application of each one depends on the context



Conceptual LCA

- Simplest form of LCA
- Basic level assessment of environmental aspects, based upon a limited and usually qualitative inventory
- The results usually indicate which components or materials have the largest environmental impacts and why
- Only help decision makers identify which products have a competitive advantage in terms of reduced environmental impacts
- Also referred as ‘Life cycle thinking’ (UNEP-DTIE, 2003)

Simplified LCA

- Simplified LCA applies the LCA method for a screening assessment covering the whole life cycle
- Uses generic data and standard modules for energy production
- A simplified assessment that focuses on the most important environmental aspects and/or stages of the life cycle
- Thorough assessment of the reliability of the results is done

Stages of Simplified LCA

Stages	Description
• Screening	Identifying those parts of the system life cycle or elementary flows that are either important or have data gaps
• Simplifying	Using the findings of the screening to focus further work on the important parts of the system or the elementary flows
• Reliability	Checking that simplifying does not significantly reduce the reliability of the overall result

Detailed LCA

- Involve the full process of undertaking LCAs
- Require extensive and in-depth, data collection
- Specifically focuses upon the target of the LCA, which if only data available generically, data must be collected specifically on the product or service under review.

Steps of detailed LCA (Consoli, et al. 1993):

- Planning
- Screening
- Data collection & data treatment
- Evaluation
- Improvement assessment

Types of LCA: Example of EV

• Conceptual	• Simplified	• Detailed
Tail pipe emissions for EVs are zero compared to substantial emissions by fossil fuel-based vehicles	General values of emissions from fuel-based vehicle manufacturing & operations are compared with generic/secondary values of emission from EV manufacturing, battery and electricity sources are compared	Exact data of all aspects of emission from mining of material up to disposal of battery or scrape are calculated and then an assessment is done for benefits of EVs over regular vehicles

Case study: An LCA comparison of last-mile distribution logistics scenarios in Milan and Turin municipalities

Source: Edoardo Croci, Matteo Donelli, Francesco Colelli, “An LCA comparison of last-mile distribution logistics scenarios in Milan and Turin municipalities”, Case Studies on Transport Policy, Volume 9, Issue 1,2021, Pages 181-190

Study area

- Study has been carried out for LTZ (Limited Traffic Zone) of Milan and Turin urban centers.
- Both cities are situated in Italy, Europe.
- Turin is an important business and cultural center
- Milan is an important historical city, second highest populated city in Italy after Rome



Image source: martinrandall.com

LTZ (Limited Traffic Zone)

- Declaring LTZ is a regulatory intervention to improve air quality, lowering noise and make area pedestrian friendly.
- Generally, it is applied in zones with high foot-fall urban centers.
- Vehicular traffic movement is allowed in limited off-peak hours or few vehicle category exempted.

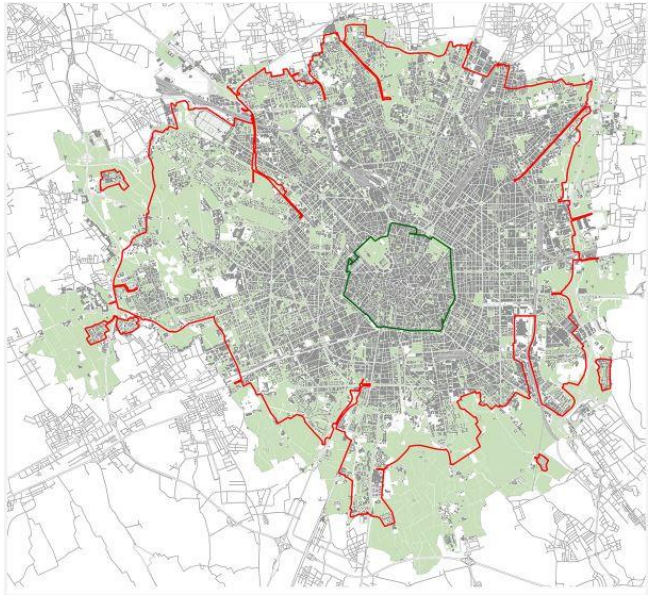


Image: Milan city boundary with LTZ as green border

Image source: urbanaccessregulations.eu

Selection of study area

Urban center of both the cities are taken for study

- Adopted in order to reduce traffic, emissions of pollutants and noise caused by vehicles in the center of urban areas through various regulatory measures such as declaring it LTZs.

- LTZs (Limited Traffic Zones) are an important area of analysis of delivery activities by logistics operators, whose operational management choices are influenced by the rules in force in the areas subject to these limitations.

- In Milan, covering an area of 8.2 square km. and that of Turin 2.58 square kilometers.

Details of Study

- Assesses the environmental footprint of last-mile distribution logistics operations through the application of a life cycle analysis method (LCA).
- The environmental impacts of the delivery activities carried out in the Limited Traffic Zones (LTZ) of the cities of Milan and Turin by a large private logistics operator are quantified.
- Through a scenario analysis, current delivery practices and an alternative solution, consisting of replacing the diesel vehicle fleet with electric vehicles (EV), are evaluated.

Steps for LCA followed

- Functional unit and the boundaries of the analyzed system
- Compilation of an inventory of the input and output items relating to the various phases of the product life cycle
- Assessment of the potential environmental impacts associated, through specific indicators
- Interpretation of the results regarding the phases of inventory analysis and the estimation of impacts in relation to the objectives of the study

Goal of study: Comparison of Scenario

- Two scenarios were considered for the LTZ of Milan and Turin
- ‘Baseline’ and ZEV/EV scenarios are compared.
- The “baseline” scenario refers to a typical day of a single large private operator and involves the exclusive use of vehicles powered by diesel fuel.
- ZEV scenario assumes diesel powered vehicle replaced by EVs

- ZEV: Zero Emission Vehicle
- EV: Electric Vehicles
- LTZ: Limited Traffic Zone

Impact category considered for LCA

Impact	Unit	Description
Climate change	Kg. CO2 eq.	Integrated heating effect in the next 100 years due to the emission of greenhouse gases into the atmosphere (CO2, CH4, N2O, etc.)
PM/smog	Kg. PM2.5 eq.	Emissions of particulate matter (PM) and its precursors (NOx, SOx, NH3)
Photochemical Ozone formation	Kg. NMVOC eq.	Production of volatile organic substances which, by the action of light, can promote an oxidation reaction that leads to the production of ozone in the troposphere
Acidification	Molc. H+ eq.	Lowering of the pH of soils, lakes, forests, due to the release into the atmosphere of acid substances, such as nitrogen oxides (NOx) and sulfur oxides (SOx), with harmful consequences on living organisms (e.g., “acid rain”)

Data collection in city of Milan & Turin

Following data has been collected:

- Types of delivery methods used (list with indication of the emission class and maximum load capacity)
- Number of vehicles used for each type
- kg of packages delivered to customers by each single vehicle in a day
- km traveled by each vehicle in a day

The period of observation from 03/07/2017 to 30/09/2017 for the city of Milan and from 21/04/2017 to 31/08/2017 for the city of Turin

Data for vehicles

Table consists following attributes:

- Vehicle classification
- Distance covered by vehicles
- Class (vehicle pollution efficiency)
- Total goods carried by vehicle

Vehicle	Distance covered (km/day)	Class	Total weight of parcels (kg)
Milan			
VOLSWAGEN-CRAFTER	64	EURO 5	1017
VOLSWAGEN-CRAFTER	61	EURO 5	1325
FORD-TRANSIT VAN	51	EURO 4	172
MERCEDES BENZ-SPRINTER	102	EURO 5	666
FORD-TRANSIT VAN	82	EURO 5	963
FORD-TRANSIT VAN	54	EURO 4	1017
PEUGEOT-BOXER	70	EURO 4	1087
VOLSWAGEN-CRAFTER	74	EURO 5	701
RENAULT-MASTER	59	EURO 4	1734
RENAULT-MASTER	58	EURO 5	1533
MERCEDES BENZ-SPRINTER	52	EURO 5	1406
FORD-TRANSIT VAN	90	EURO 5	777
MERCEDES BENZ-SPRINTER	45	EURO 5	62
RENAULT-MASTER	59	EURO 5	1232
Turin			
FORD-TRANSIT VAN	107	EURO 5	782
FIAT DOBLO'	61	EURO 5	1063
FORD-TRANSIT VAN	91	EURO 5	3289
FIAT DOBLO'	67	EURO 5	468

Ecoinvent input data on load/carrying capacity

- Ecoinvent: World's leading LCI (Life Cycle Inventory) database for products with transparency & consistency
- Avg. Load: Avg. weight carrying of any vehicle
- GVW: Gross vehicle weight
- Load factor: Ratio of the average load to total vehicle freight capacity

Lorry size class (Ton)	Avg. load (Ton)	GVW (Ton)
3.5-7.5	0.98	4.98
7.5-16	3.29	9.29
>32	15.96	29.996

Input data on diesel vehicles tested in Milan and Turin

Vehicle Gross Vehicle Weight Rating (GVWR)	Max capacity	Average load factor	Utilization rate	Fuel consumption (kg/tkm)
Lorry 3.5–7.5 metric ton	4.98 t	0.98 t	20%	0,1093
Van Type 1	1,25 t	0,625 t	50%	0,2222
Van Type 2	1 t	0,5 t	50%	0,25

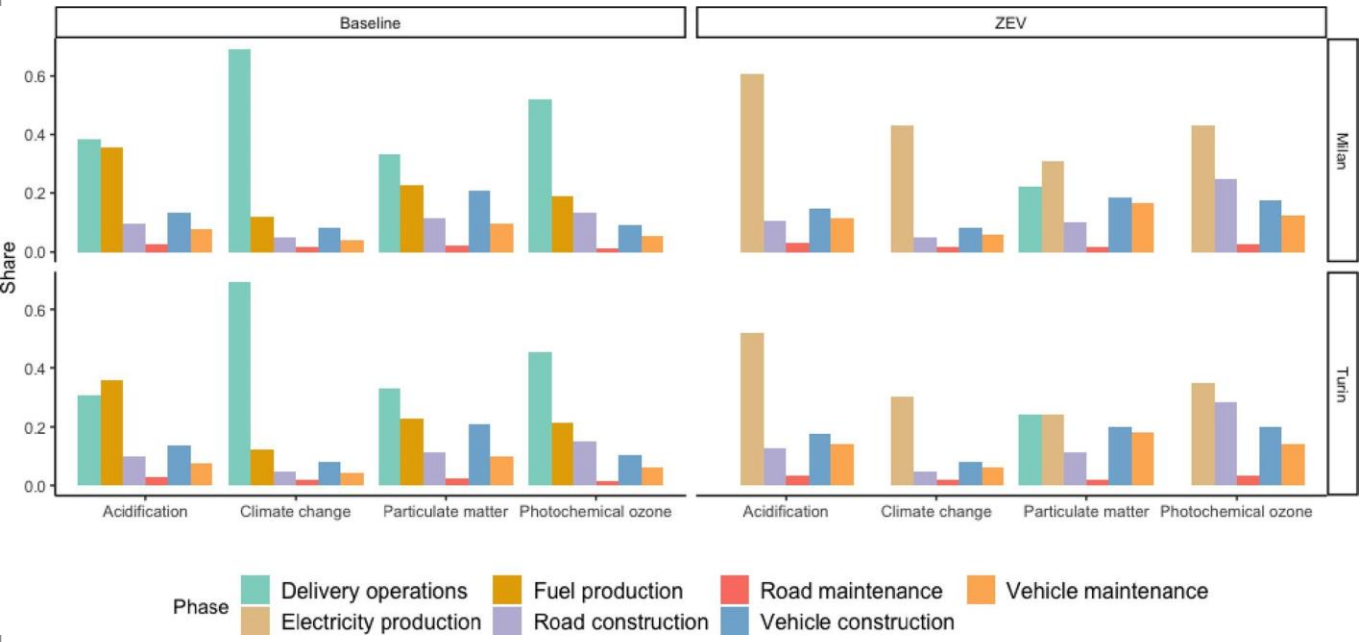
Input data on electric vehicles

Vehicle	Max capacity	Max battery capacity (kwh)	Max distance (km)
Light	730 kg	24	170
Heavy	2000 kg	28 (1–3 batteries)	280 (3 batteries)

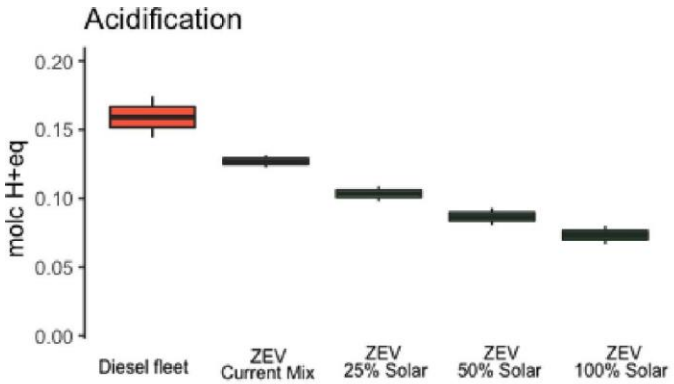
Results : Impacts categories

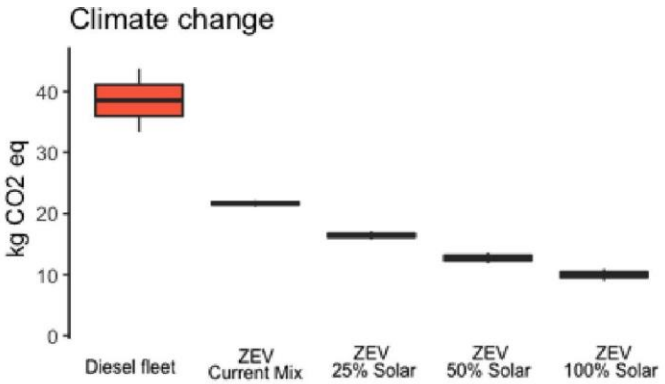
	Baseline		ZEV		
	Milan				
	Per ton of parcels	Daily total	Per ton of parcels	Daily total	Share of ZEV over Baseline
Climate change (kg CO2 eq)	33	457	21	290	–37%
Particulate matter (kg PM2.5 eq)	0,013	0,18	0,015	0,20	13%
Photochemical ozone formation (kg NMVOC eq)	0,135	1,85	0,072	0,98	–47%
Acidification (molc H + eq)	0,144	1,97	0,122	1,67	–15%
	Turin				
Climate change	44	245	22,17	124	–49%
Particulate matter	0,017	0,1	0,018	0,1	4%
Photochemical ozone formation	0,157	0,9	0,082	0,5	–48%
Acidification	0,174	1,0	0,131	0,7	–24%

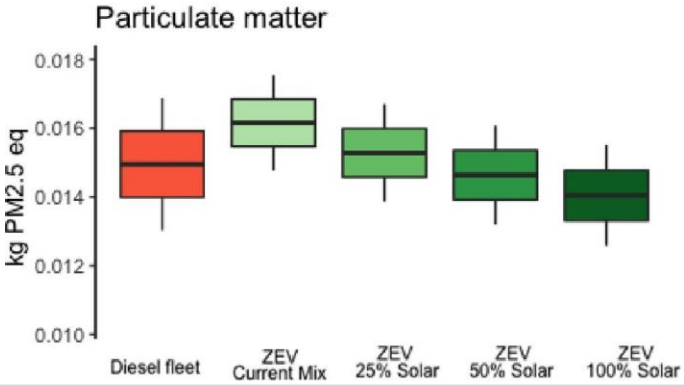
Results: Comparative effects on impact categories

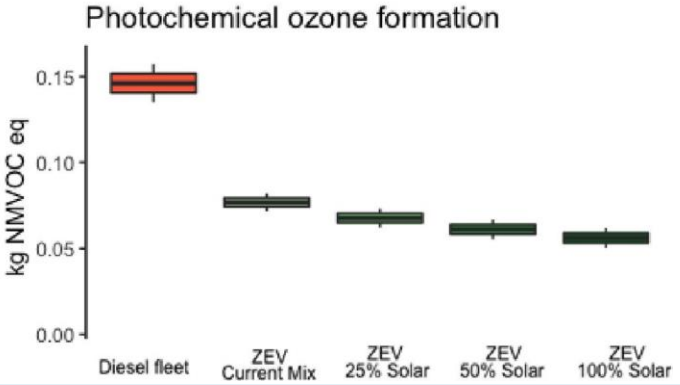


Results: Comparative effects of energy sources on emission









Difference in Results for Milan & Turin

Ideally results should be same for both the cities.

Differs due to the following reasons:

- Composition of the fleet in Turin is exclusively of EURO 5 vehicles, while in Milan combination of EURO 4 and EURO 5 vehicles

- Diversity of the routes that the vehicles make in the two LTZ to complete the same load of deliveries.

- Different composition of electric vehicles that for the Turin area also include the use of heavy ZEV vehicles, which has lower specific consumption per kg of goods transported

Conclusion of study 1/2

- The adoption of the ZEV fleet cuts GHG by 37% in Milan and by 50% in Turin.
- Adoption of the ZEV fleet combined with the current electricity mix results in an increase in the particulate matter emissions of 13% in Milan and 4% in Turin.
- Adoption of the ZEV fleet cuts NMVOC emissions by almost 50% in both cities.

Conclusions for study 2/2

- Our analysis confirms that the adoption of the “ZEV” fleet should be combined with other policies favoring the transition to a decarbonized power system in order to fully capture the environmental benefits which can be associated to the electric fleet

- Sourcing 50% of the electricity used by the ZEV from dedicated photovoltaic columns would make both scenarios’ emissions comparable, while sourcing 100% of it from solar power would result in a reduction in the particulate emissions of −7%. (However, it needs to be looked with local vs regional emission perspectives and that way impacts will be different).

Conclusions

- LCA should be done according to ISO standards.
- Various LCAs are done as per requirement of study and its objective.
- LCA can be used efficiently for scenario comparison which can be applied in various cases.

References

- Asia-Pacific Economic Corporation; 2004, “Life Cycle Assessment: Best practices adopted in ISO 14040 series”

- Department of Environmental Affairs and Tourism, South Africa; 2004 “Life cycle assessment”

- Edoardo Croci, Matteo Donelli, Francesco Colelli, 2021 “An LCA comparison of last-mile distribution logistics scenarios in Milan and Turin municipalities”, Case Studies on Transport Policy, Volume 9, Issue 1, Pages 181-190

Thank You